radleys
accelerating chemistry

inkarp®
Your Knowledge... Our Solution...!

International Product Guide 2024

Equipment for chemical synthesis, process development, evaporation and work-up

Accelerating chemistry

Mya 4™ Reaction Station

One reaction station with limitless possibilities

A 4-zone reaction station offering safe and precise heating, active cooling, software control and data-logging for 24/7 unattended chemistry

Features

- 4 Independent zones - each with heating and active cooling
- -30 °C to +180 °C
- 2 ml to 400 ml vessels - wide range of styles
- Magnetic and overhead stirring
- Software - control and log results automatically
- Control 3rd party devices

Safer, cleaner, greener and more productive

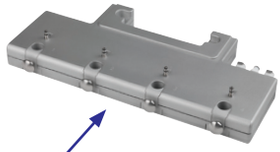
- Replace inefficient, messy and unsafe oil and ice baths
- Save space compared with separate reaction set-ups
- Software control improves safety, reduces manual errors, and allows 24/7 unattended chemistry, for increased productivity
- Create, repeat and share experiments and results with ease and accuracy
- Easily manage complex multi-step and multi-device experiments
- Integrate and link 3rd party devices such as syringe pumps and pH sensors

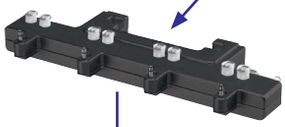
Touch-screen Control Pad

Wide range of vessel styles and sizes
- 2 ml to 400 ml vessels
- Vials, tubes, round bottom flask and straight sided process vessels

Mya Compact Stirrer
- Stirs from 100 to 1000 rpm
- Control each position independently
- Powerful, high torque motor
- Tool-free attachment

Reflux Head
- Water-cooled
- Efficient refluxing
- Distributes inert gas





Manifold Head
- Distributes water for glass condensers
- Distributes inert gas

Excellent visibility
- Easily view your vessel contents

Up to 200 °C difference in adjacent zones
- High performance insulation minimises zone cross-talk

Temperature -30 °C to +180 °C
- Independent control of each zone
- Block or solution control
- Pt100 temperature probes

Space saving
- Compact benchtop footprint
- Less space than 4 separate reaction set-ups

Cool to -25 °C with tap water
- No need for a separate chiller
- Powerful Peltier technology provides rapid cooling

Magnetic stirring from 100 to 1000 rpm
- Control each position independently

Touch-screen Control Pad
- Supplied as standard
- Intuitive and easy to use
- Set automated profiles or use manual control
- Compact design with 10" display

Single or multi-user operation
- Control heating/cooling and stirring of each zone independently

Optional PC Control Software
- Integrate and control 3rd party devices such as pumps, balances and pH sensors

Configure Mya 4 in the way that suits your chemistry

A flexible tool for a wide range of applications, from discovery chemistry to process development

Applications:

- Process development
- Design of Experiment (DoE)
- Biopharma research
- Scale up
- Route scouting
- Crystallisation studies
- Polymorph screening
- Lead optimisation
- Reaction optimisation
- Reagent, catalyst and solvent screening

Mix & Match

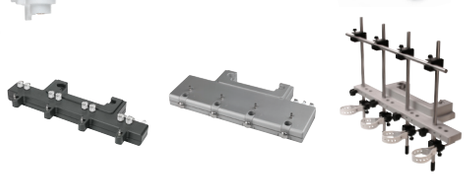
One reaction station with a wide range of options and accessories to fit your needs

Stirring options

Overhead / Magnetic stirring

Head options

Manifold / Reflux / Support Head



Multi-neck lids or Reflux tubes

Glass or PTFE lids
Standard or wide-neck reflux tubes

Vessel styles and volumes

Tubes, vials, RBFs and Process vessels from 2 ml to 400 ml

Aluminium inserts

For each vessel size

Select accessories

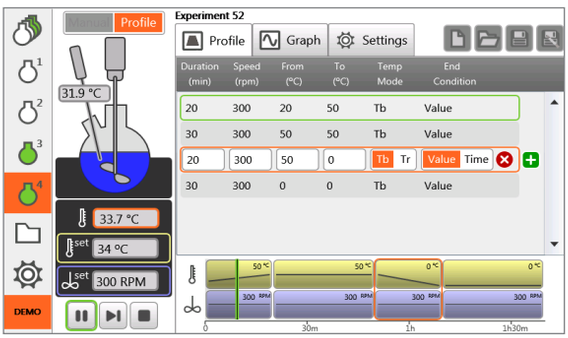
Temperature probes, condensers, funnels, storage solutions and many more

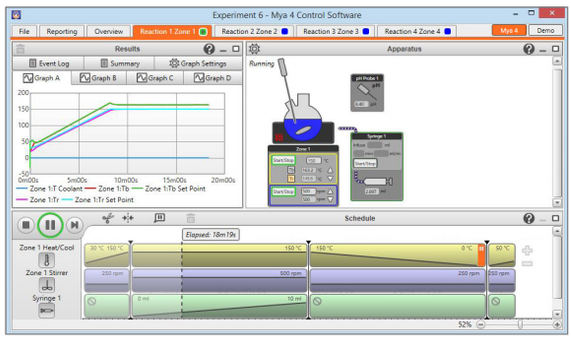
Touch-screen Control Pad

- Supplied as standard with Mya 4
- Intuitive and easy to use
- Compact footprint
- Set automated profiles or use manual control
- 10” display

PC Control Software - optional

- Integrate and control 3rd party devices
- Create complex experiments with any number of steps
- Report Wizard creates reports in rich text format or export results in CSV

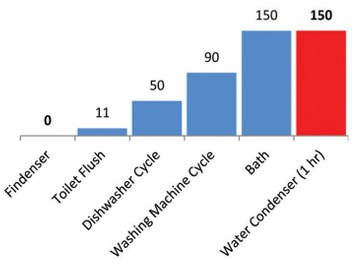




What are the benefits?

- No risk of flooding from running water
- Eliminate the cost of water purchase and disposal
- For solvent volumes from 5 ml up to 1 litre
- Helps meet sustainable water reduction targets

Average water consumption of common household appliances compared to a Findenser and water condenser



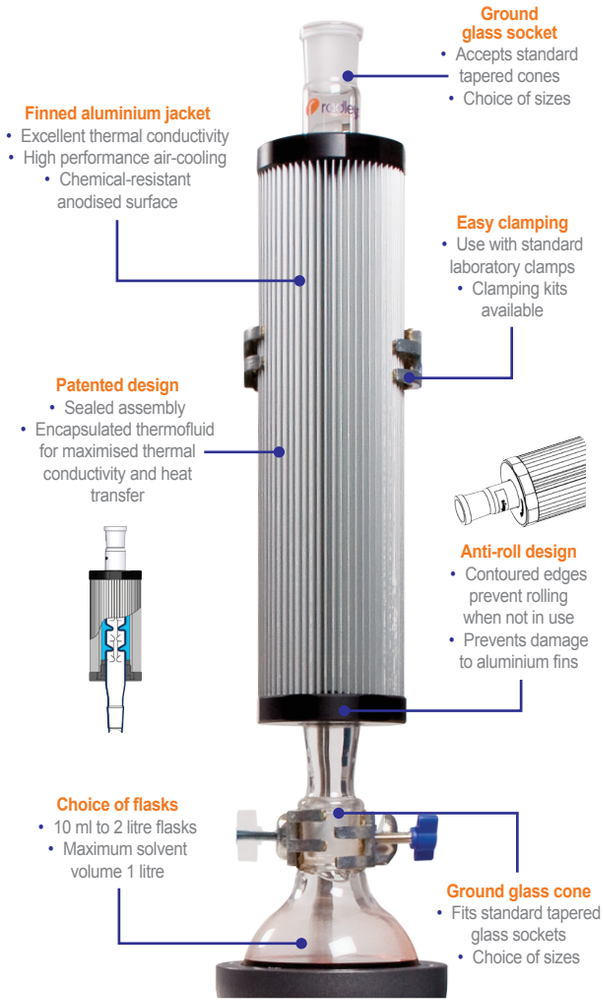
Choice of B14, B19, B24 and B29 joint sizes

Findenser™ - *prevents flooding and saves water*

Replaces water-cooled condensers in over 95% of common chemistry applications

How does Findenser work?

- Findenser comprises an internal glass condenser and an external, finned aluminium jacket, between which a small amount of water is permanently sealed.
- The glass condenser design has a greater internal surface area than traditional air condensers, increasing heat transfer capacity.
- The finned jacket fits around the glass condenser, further increasing the external surface area.
- The result is a 'SUPER air condenser'.



Findenser requires no running water to operate. Water is a precious resource. It makes little economic or environmental sense to waste thousands of litres just to cool a single condenser.

Performance testing

A range of solvents, in identical flasks and set-ups, were tested with a Findenser, water condenser and air condenser to record solvent loss by weight.

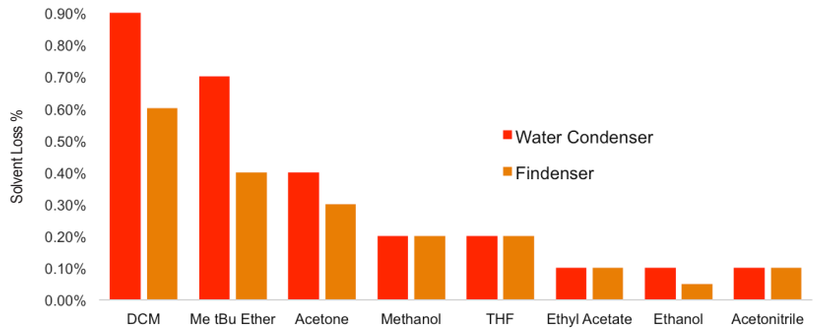
Findenser compared to an air condenser

For synthesis with low boiling point solvents, Findenser showed a significant improvement in solvent retention. With acetone or DCM the reaction boiled dry when using an air condenser, yet Findenser retained 95% of the solvent under the same conditions.

For synthesis with medium boiling point solvents, Findenser delivered improved solvent retention particularly with larger volumes and high temperatures.

Findenser compared with a water condenser

Under identical conditions, Findenser retained solvent to the same level (within the limits of experimental error) as a water condenser with a range of solvents. Diethyl ether is not recommended for use with Findenser.



Heat-On™ Block System - *the safer alternative to oil baths*

The safest, fastest and most efficient way to heat and stir round bottom flasks from 10 ml to 5 litres

Features

- Replace messy oil baths and heating mantles and avoid spills.
- Make your chemistry safer, cleaner and faster.
- Solid aluminium blocks provide even heating.
- Lightweight design allows rapid heating.
- Uniquely shaped well design eliminates cracking of flasks.
- Blocks feature two probe holes and optional lifting handles.
- Use up to 260 ˚C.
- Also available for Florentine flasks, vials and tubes.

Heat-On Multi-Well Block with 50 ml and 100 ml flask inserts

The risk of oil fires and injury from hot oil spills, plus the mess and cost associated with the use of oil, means that oil baths no longer represent safe working practice in labs. Heating mantles are expensive, difficult to clean, do not respond well to spills and often create hot spots when heating. Scientists are increasingly turning to specially designed aluminium blocks located on stirring hotplates to heat standard round bottom flasks.

Not all block designs are the same

Test results show that Heat-On heats up to 66% faster and uses 30% less energy than other brands of block.

Visit radleys.com to download the application bulletin

260˚C

Heat-On Multi-Well Block
- Accepts two 50 ml or 100 ml flasks, or one 150 ml flask

Over 50 styles a sizes to choose f
- Accepts tubes and
- 1 ml to 5 litre

Two temperature probe holes
- Accept 3 mm ø probe

Lightweight design enables rapid heating
- Use up to 260 ˚C
- Uses 30% less energy
- Heats water 66% faster

Unique well design prevents flask cracking
- Unlike many other inferior blocks, Heat-On will not crack your flask when cooling

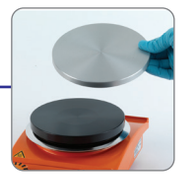
Optional lifting handles
- Quick-release mechanism for safe removal

Anodised finish
- Heat-On blocks are also available with a lower cost anodised finish if preferred

Fluoropolymer coating
- Superb chemical resistance
- Easy to clean
- Speeds up heating times

PTFE safety covers reduce the risk of burns
- Reduces surface temp. by up to 50%
- Reduces energy consumption by 15%
- Available for the most popular Heat-On sizes

Square hotplate adapter
- Allows Heat-On to sit on a square top plate up to 200 x 200 mm



Fits all leading hotplate brands
- Suitable for 145 mm ø top plates
- Optional adapter for 135 mm ø top plates

Cool-It™ Bowl - *the virtually unbreakable dewar*

The safe and efficient way of cooling and stirring round bottom flasks to -78 ˚C

Cool-It replaces fragile glass dewars, unstable plastic bowls and keeps your chemistry colder for longer. The compact and virtually unbreakable Cool-It insulated bowls are designed to fit onto a standard stirring hotplate to cool and stir round bottom flasks, beakers and test tubes etc.

Cool-It keeps it cooler for longer

-78˚C

- Cool-It will keep your sample below -70 °C for up to 5 times longer than a plastic bowl.
- Cool-It will keep your reactions below -70 °C for twice as long as a glass dewar.

Two part lid be removed with flask in situ

Small Cool-It bowl for flasks up to 400 ml

Large Cool-It bowl for flasks up to 2 litres

Cool-It accessories
- Clamps, stand, digital thermometer, scoop, protective gloves, apron and face shield

Lid improves cooling
- Keeps reaction cool for up to 20% longer
- Easily fitted once your flask is in place
- Minimises ice formation on flask
- Prevents spitting

Virtually unbreakable
- Manufactured from a robust, chemically-resistant HDPE
- High quality insulated foam core
- Unlike fragile glass dewars is virtually unbreakable

Protects your stirrer and minimises spills
- Cool-It minimises frost on the outer surfaces, protecting your stirrer from moisture ingress
- Cool-It fits securely on the top plate minimising spills and the risk of the bowl being knocked off the top plate

Fits on all popular brands of stirring hotplate
- Suitable for use with top plates of 135 mm and 145 mm diameter

Easy pour spout and handle
- Non-drip spout and handle makes the disposal of solvents safer and easier
- Avoid spills for a safer working environment

Cool-It keeps it cooler for longer
- Cool-It will keep your sample below -70 °C for up to 5 times longer than a plastic bowl
- Cool-It will keep your reactions below -70 °C for twice as long as a glass dewar

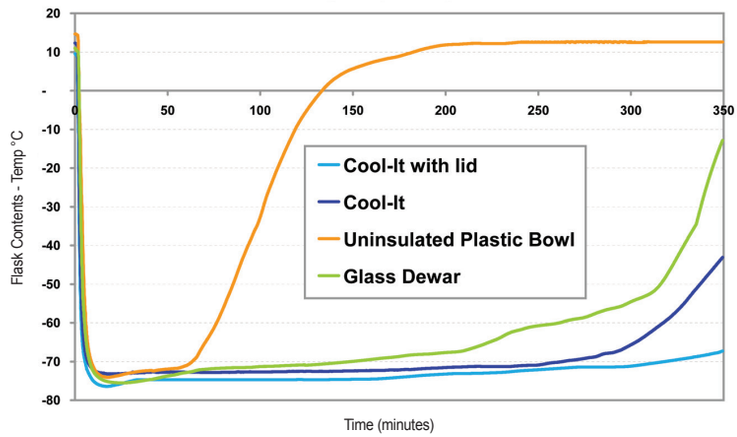
Large and small bowl options
- Small for flasks up to 400 ml
- Large for flasks up to 2 litres

Cool-It accepts all shapes and sizes of round bottom and Florentine flask up to 2 litres

Easy pour design with spout and handle

Cool-It vs. Glass Dewar

Acetone and dry ice, 250 ml flask, 100 ml acetone



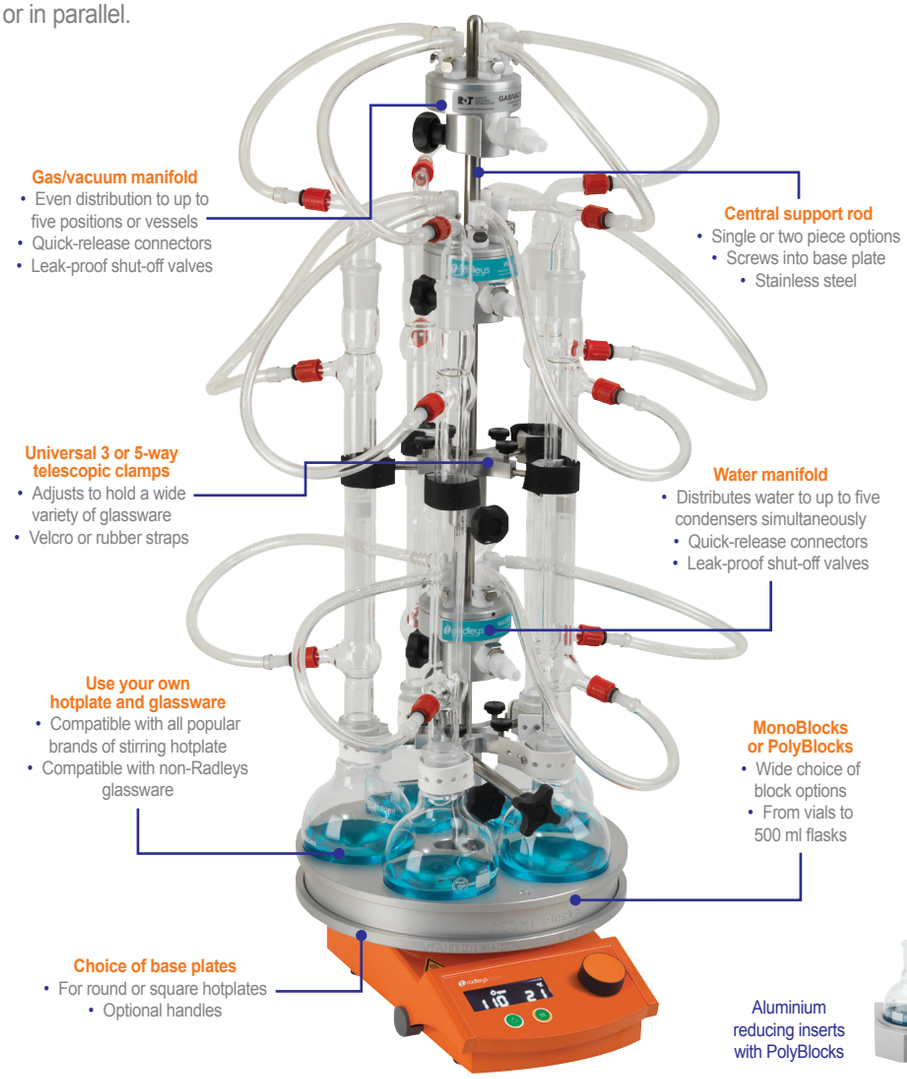
StarFish™ Multi-Experiment Work Station - *the space saver*

StarFish is a modular, general purpose heating and stirring work station

Whether you want to just heat and stir or perform more complex parallel experiments, StarFish really can make your life easier and improve productivity.

Features

- Fits all leading brands of stirring hotplate.
- Accepts vials, test tubes and round bottom flasks.
- Set up vessels individually or in parallel.



Gas/vacuum manifold
- Even distribution to up to five positions or vessels
- Quick-release connectors
- Leak-proof shut-off valves

Central support rod
- Single or two piece options
- Screws into base plate
- Stainless steel

Universal 3 or 5-way telescopic clamps
- Adjusts to hold a wide variety of glassware
- Velcro or rubber straps

Water manifold
- Distributes water to up to five condensers simultaneously
- Quick-release connectors
- Leak-proof shut-off valves

Use your own hotplate and glassware
- Compatible with all popular brands of stirring hotplate
- Compatible with non-Radleys glassware

MonoBlocks or PolyBlocks
- Wide choice of block options
- From vials to 500 ml flasks

Choice of base plates
- For round or square hotplates
- Optional handles

Applications

- Heating and stirring
- Synthesis
- Distillation
- Extraction
- Digestion
- Concentration

Space saving
Uses less space than multiple heating and stirring set-ups.

Increases productivity
Multiple positions allow you to heat, stir and reflux experiments in parallel.

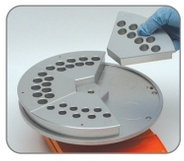
Cost effective
Use your existing stirring hotplate and glassware. Eliminate the cost of multiple set-ups.

Safer, cleaner working
Eliminate oil baths, reduce spills, mess and accidents.

Flexible
Use as many positions as you want.

Easy to use
Easy to store and quick to assemble.

Compact
Store spare components in a drawer and not on the bench.


PolyBlock

MonoBlock

Aluminium reducing inserts with PolyBlocks

Mix 'n' match the components you need, when you need them

Simple and convenient, the Carousel Work-Up Station will reduce post-synthesis bottlenecks.

Work-Up Station for parallel or sequential work-up of 12 samples, using filtration, phase separation, liquid/liquid extraction or SPE.

Carousel 12 Plus Reaction Station™

The patented Carousel 12 Plus simultaneously heats/cools, stirs and refluxes multiple samples under an inert atmosphere

An effective personal synthesis station for parallel solution phase chemistry and solid supported reagent based synthesis.

220˚C

Features

- Accepts up to 12 glass tubes with a reaction volume of 1 ml to 20 ml.
- Powerful, even stirring - fits onto a Carousel Stirring Hotplate.
- Rapid heating to 220 ˚C and cooling to -78 ˚C.
- Quick to set up and easy to use.
- Easy viewing of tube contents during experiments.
- Removable water-cooled reflux head.
- Perform reactions under an inert atmosphere.
- Fluoropolymer coating for chemical resistance and easy cleaning.
- PTFE caps feature a quick-thread for fast attachment to glass tubes.
- Removable reflux head allows reaction tubes to be transferred between heated base, cooled base or stand.

The world's most popular parallel synthesiser

Heated directly by the stirring hotplate with optional digital control and Pt1000 temperature sensor.

Quick-release inlet/outlet for vacuum and gas, combined with a radial gas distribution system and gas-tight caps, allow reactions under an inert atmosphere.

Chemical-resistant Easy-On PTFE caps feature a quick-thread for fast attachment to the glass tubes and push-on connections to the s/steel gas outlets.

Water-cooled aluminium reflux head provides efficient refluxing within individual reaction tubes

Quick-release water couplings with cut-off valves for ease of connection to cooling water supply.

Accepts up to 12 tubes with reaction volume of 5 to 20 ml (1 ml with reduced volume tubes).

Fluoropolymer coating for chemical resistance and easy cleaning.

Easy to operate and set up with minimal training. No electrical or moving parts ensure maintenance-free operation.

Unique removable fluoropolymer insulation plate helps insulate the heated base for faster heating and energy savings of up to 36%.

Base design improves heat transfer and provides energy savings. Maximum operating temperature 180 ˚C (220 ˚C for short periods).

Utilises the single rotating magnetic field of the hotplate stirrer to stir all the positions evenly and powerfully.

Visibility slots allow easier viewing of tube contents. Easy to rotate round design gives access to all tubes with no need to lean into the fume cupboard.

Rare earth cross shaped stirring bars for vigorous stirring and a deeper vortex, without jamming.

Carousel Stirring Hotplate offers higher temperature, powerful stirring and digital control.

Control and log your chemistry with AVA Lab Control Software

ava lab control software

Cooled Carousel 12 Plus Reaction Station™

Cost effective, low temperature parallel synthesis down to -78 ˚C

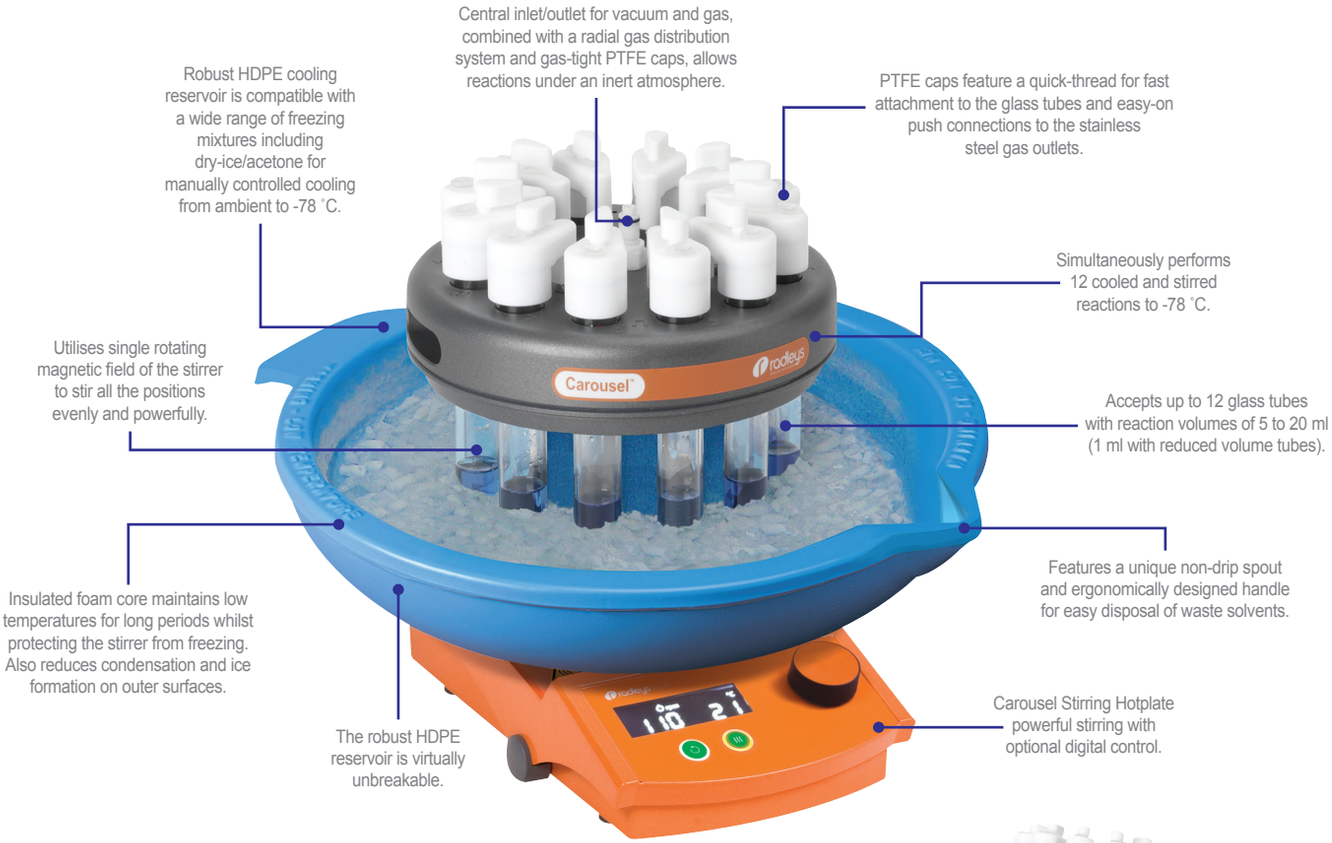
The innovative Cooled Carousel 12 Plus reservoir is designed to accept the removable reflux/inerting head from the Carousel 12 Plus, allowing reaction tubes to be easily and rapidly transferred between heating and cooling bases.

Features

- Simultaneously performs 12 cooled and stirred reactions to -78 ˚C.
- Powerful, even stirring - reservoir fits on to a Carousel Stirring Hotplate.
- Robust HDPE cooling reservoir is compatible with a wide range of cooling mixtures, including dry-ice/acetone for manually controlled cooling from ambient down to -78 ˚C.
- Features a non-drip spout and handle for disposal of waste solvents.
- Insulated foam core maintains low temperatures for long periods whilst protecting the stirrer from freezing. Also reduces condensation and ice formation on outer surfaces.
- HDPE lid keeps your reaction cooler for longer, minimises ice formation on your tubes (maintaining visibility of the contents) and prevents spitting from the cooling mixture.
- The robust HDPE reservoir is virtually unbreakable.

Upgrade your Carousel to perform cooled chemistry

Transfer the reflux/inerting head to the cooled reservoir



Carousel 12 Plus Stand

The Carousel stand is designed to support the reflux/inerting head either with or without reaction tubes.

The heavy duty metal stand is fluoropolymer coated for improved chemical resistance and ease of cleaning. The integral drip tray catches any dripping condensation from tubes and gives excellent stability.

The Tornado integrates with the Carousel 6 Plus to provide powerful, controlled mechanical stirring of up to six round bottom flasks.

Offers unrivalled stirring for both viscous samples and for the dispersion of delicate solids in solution.

See page 16

Carousel 6 Plus Reaction Station™

The patented Carousel 6 Plus simultaneously heats, stirs and refluxes multiple samples under an inert atmosphere

Accepts round bottom flasks: 5 ml, 10 ml, 25 ml, 50 ml, 100 ml, 170 ml and 250 ml sizes.

Features

- Powerful, even stirring - fits onto a Carousel Stirring Hotplate.
- Rapid heating to 180 ˚C.
- Quick to set up and easy to use.
- Water-cooled reflux head.
- Perform reactions under an inert atmosphere.
- Easy viewing of flask contents during experiments.
- 100 ml and 250 ml azeotropic (Dean and Stark) flask option.
- PTFE caps feature a quick-thread for fast attachment to flasks.

A wide range of accessories including liquid additions funnels, powder funnels and rotary evaporator adapters

Central inlet/outlet for vacuum and gas, combined with a radial gas distribution system and gas-tight PTFE caps, allow reactions under an inert atmosphere

Heated directly by the stirring hotplate with optional digital control and Pt1000 temperature sensor

Round design makes all reaction flasks visible, with no need to lean into the fume hood

Chemical-resistant Easy-On PTFE caps feature a quick-thread for fast attachment to the glass tubes and push-on connections to the stainless steel gas outlets

Water-cooled aluminium reflux head provides efficient refluxing within individual glass reaction flasks. Quick-release couplings prevent water loss during set-up/breakdown

Aluminium inserts allow easy removal of flasks and good temperature transfer for refluxing

Range of glass vessels: 5 ml, 10 ml, 25 ml, 50 ml, 100 ml, 170 ml and 250 ml round bottom flasks, as well as vessels with one or two sidearms

PTFE heat protection ring protects user from contact with hot base

Rare earth elliptical PTFE stirring bar provides powerful stirring and a deep vortex

Control and log your chemistry with AVA Lab Control Software

ava lab control software

Utilises the single rotating magnetic field of the hotplate stirrer to stir all the positions evenly and powerfully

Round aluminium base transmits heating evenly to all positions. Compact size has small bench-top footprint. Easy to store

Carousel Stirring Hotplate offers higher temperature, powerful stirring and digital control

Aluminium inserts for 5 ml, 10 ml, 25 ml, 50 ml, 100 ml and 170 ml flasks

5 ml Reaction Flask Reflux Tube & PTFE Cap

10 ml Reaction Flask Reflux Tube & PTFE Cap

25 ml Reaction Flask Reflux Tube & PTFE Cap

50 ml Reaction Flask Reflux Tube & PTFE Cap

50 ml Flask with Sidearm Reflux Tube & PTFE Cap

100 ml Reaction Flask Reflux Tube & PTFE Cap

Cooled Carousel 6 Plus Reaction Station™

Cost effective low temperature parallel synthesis down to -78 ˚C

The Cooled Carousel 6 Plus allows chemists to perform sub-ambient reactions in a range of flasks from 5 ml to 250 ml with the option of an inert, moisture-free atmosphere.

Features

-78˚C

- Simultaneously perform up to six cooled and stirred reactions to -78 ˚C.
- Powerful, even stirring - reservoir fits on to a Carousel Stirring Hotplate.
- Robust HDPE cooling reservoir is compatible with a wide range of cooling mixtures, including dry-ice/acetone for manually controlled cooling from ambient down to -78 ˚C.
- Insulated foam core maintains low temperatures for long periods, whilst protecting the stirrer from freezing. Also reduces condensation and ice formation on outer surfaces.
- HDPE lid keeps reactions cooler for longer, minimises ice formation on flasks (maintaining visibility of the contents) and prevents spitting from the cooling mixture.
- Round design makes all reaction flasks visible and allows easy addition of reagents and solvents, with no need to lean into the fume hood.

Visit www.radleys.com to download a PDF on the Cooled Carousel 6 Plus

HDPE lid reduces frost formation and reduces the risk of solvents spitting.

Central inlet/outlet for vacuum and gas, combined with a radial gas distribution system and gas-tight PTFE caps, allow reactions under an inert atmosphere

Chemical-resistant Easy-On PTFE caps feature a quick-thread for fast attachment to the glass tubes and push-on connections to the s/steel gas outlets

Round design makes all reaction flasks visible, with no need to lean into the fume cupboard

Robust HDPE cooling reservoir is compatible with a wide range of freezing mixtures including dry-ice/acetone for manually controlled cooling from ambient to -78 ˚C

Insulated foam core maintains low temperatures for long periods whilst protecting the stirrer from freezing and also reduces condensation and ice formation on outer surfaces

Rare earth elliptical PTFE stirring bar provides powerful stirring and a deep vortex

Utilises the single rotating magnetic field of the hotplate stirrer to stir all the positions evenly and powerfully

No electrical or moving parts ensures maintenance free operation. Easy to set up with minimal training time

Carousel Stirring Hotplate powerful stirring with optional digital control

Fits on a standard Carousel Stirring Hotplate

Compact size has a small bench-top footprint and is easy to store

100 ml Flask with Sidearm Reflux Tube and PTFE Cap

170 ml Reaction Flask Reflux Tube and PTFE Cap

250 ml Reaction Flask Reflux Tube and PTFE Cap

250 ml Wide Neck Vessel

250 ml Flask with 2 Sidearms with Dropping Funnel

250 ml Reaction Flask with Dropping Funnel

250 ml Azeotropic Reaction Flask with Dropping Funnel

Tornado with 250 ml wide neck flasks

Tornado™ Overhead Stirring System

Use a single overhead stirrer to stir up to six round bottom flasks from 50 ml to 250 ml simultaneously. Increase your stirring productivity by up to 600%

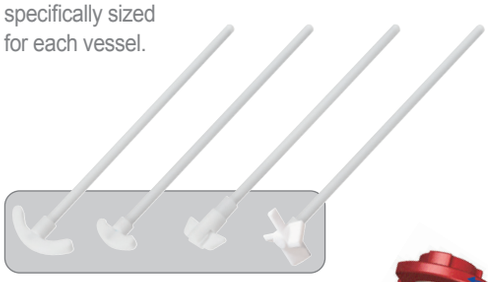
The Tornado allows powerful, controlled mechanical stirring of round bottom flasks with the Carousel 6 Plus Reaction Station, offering unrivalled stirring for both viscous samples and for the dispersion of delicate solids in solution.

Features

- Integrates with Carousel 6 Plus to provide heated and stirred reactions.
- Rapid heating to 180 ˚C with water-cooled reflux head.
- Perform reactions under an inert atmosphere.
- Accepts 50 ml, 100 ml and 250 ml round bottom flasks.
- Uses a single overhead stirrer - save space and money compared to multiple set-ups.
- Compatible with all leading brands of overhead stirrer.
- 2-speed drive allows overhead stirrers with less torque to be used for higher viscosities.
- Stir to 1000 rpm in low viscosity.
- Max. viscosity 10000 mPas at 500 rpm.

PTFE stirring shafts

Choice of centrifugal, anchor and propeller PTFE stirrers, specifically sized for each vessel.



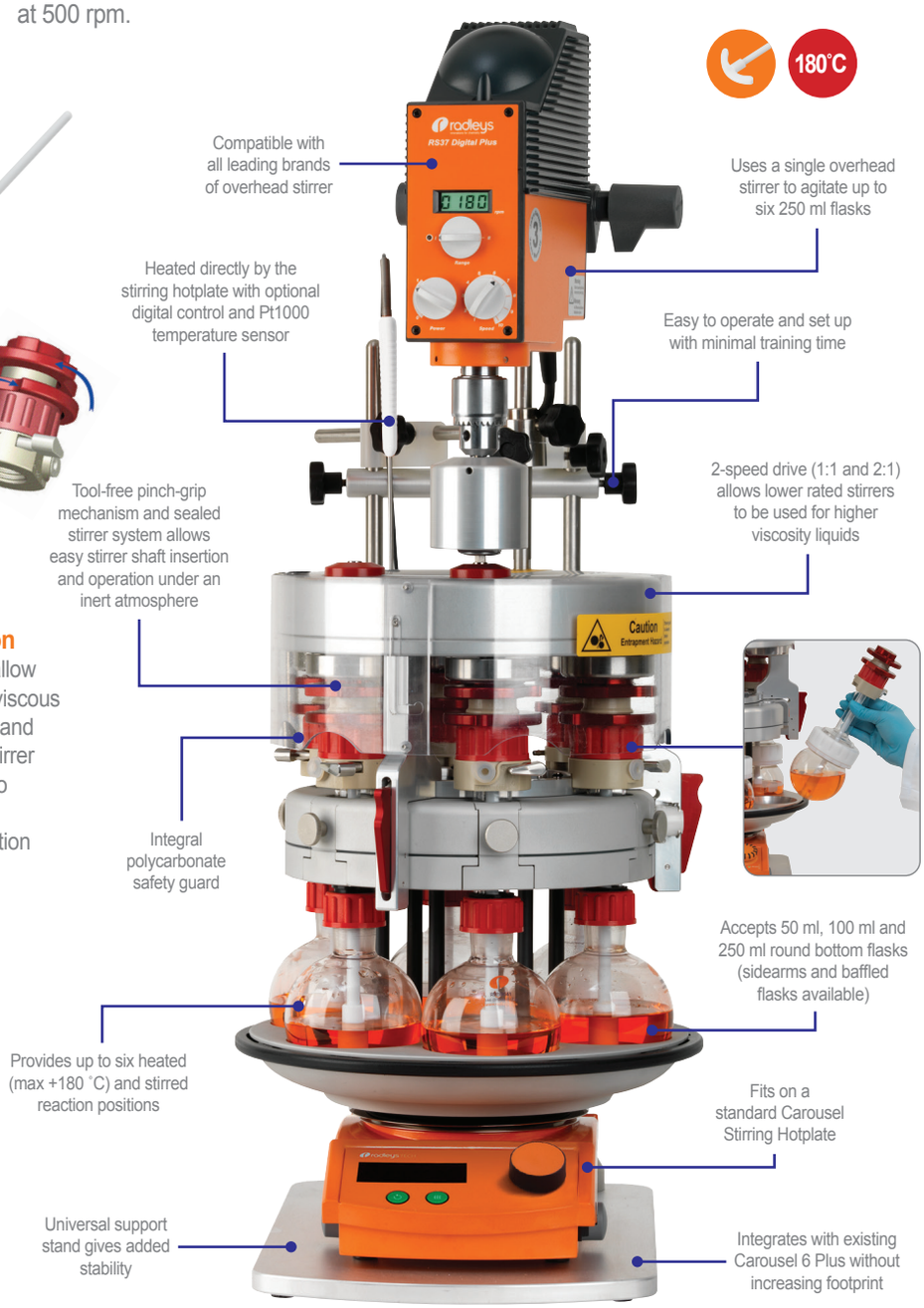
Choice of flask sizes and styles with wide neck option

Wide neck flasks allow easier removal of viscous and solid samples and the use of larger stirrer blades. Baffles also improve stirring by disturbing the creation of a vortex.

Azeotropic vessels

250 ml azeotropic (Dean and Stark) flask option.

250 ml azeotropic vessel with dropping funnel and centrifugal PTFE stirrer



GreenHouse Work-Up provides rapid sequential and parallel purification in a 24 well MTP footprint using standard filtration, phase separation and SPE columns.

Designed to make your parallel chemistry work-up and purification quick and easy. See page 20.

GreenHouse Plus Parallel Synthesiser™

The GreenHouse Plus provides 24 heated and stirred glass reactions with volumes from 0.5 ml to 7 ml. The combined reflux and additions head allows for convenient additions or withdrawals whilst maintaining an inert atmosphere

GreenHouse Plus brings all the benefits in productivity of parallel synthesis at a fraction of the cost of automated systems. Holding 24 glass reaction tubes in a removable reaction block with the same footprint as a standard microtiter plate (MTP), the GreenHouse Plus facilitates rapid transfer of samples by multi-channel pipettor or robotic systems.

Features

- Powerful stirring and rapid heating to 150 ˚C.
- Removable water-cooled reflux head.
- Perform reactions under an inert atmosphere.
- Easy viewing of tube contents during experiments.
- Nickel-plated aluminium offers excellent chemical resistance.

Designed for the synthesis of small compound libraries and drug discovery

Water-cooled aluminium reflux head with nickel condensing fingers provides efficient refluxing within individual glass reaction tubes.

Heated directly by the stirring hotplate with optional digital control and Pt1000 temperature sensor.

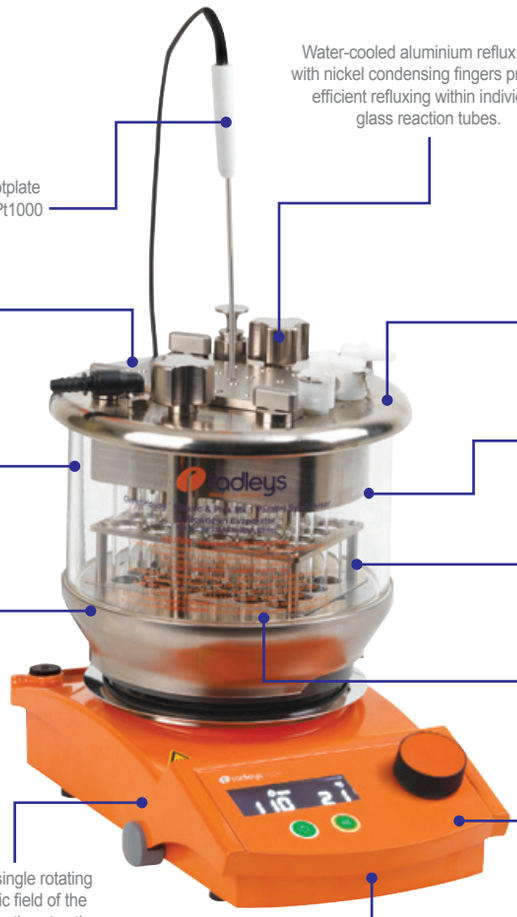
Can be converted into GreenHouse Blowdown Evaporator

A choice of septum mats are available.

Combined reflux and additions head allowing for convenient additions or withdrawals whilst refluxing.

Cylindrical glass gas enclosure provides visibility of all 24 reaction tubes with no need to lean into the fume hood

No electrical or moving parts ensures maintenance-free operation. Easy to operate and set up.



Round aluminium base transmits heat evenly to all positions. Compact size has small bench-top footprint and is easy to store.

Provides 24 heated and stirred glass reaction positions with a reaction volume of 0.5 to 7 ml.

'V-Mag' technology uses a vertically positioned stirring bar to maximise the uniformity of the stirring within each reaction tube.

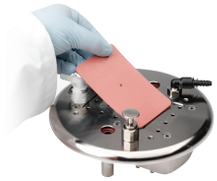
Control and log your chemistry with AVA Lab Control Software

Utilises single rotating magnetic field of the hotplate stirrer to stir all 24 positions

Fits on a standard Carousel Stirring Hotplate

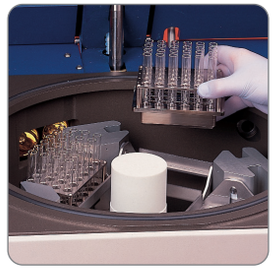
Carousel Stirring Hotplate offers higher temperature, more powerful stirring and digital control

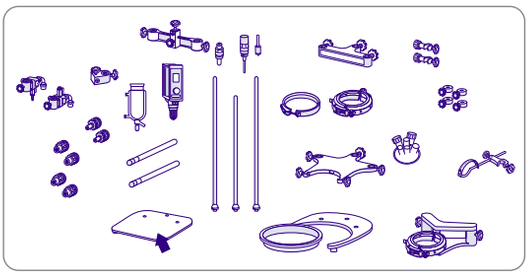
Combined reflux and additions head with nickel condensing fingers



GreenHouse Plus allows additions and withdrawals through sealing mats

Reaction block fits directly into Genevac vacuum centrifuges





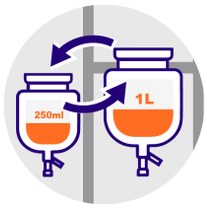
Reactor-Ready Flex™ Lab Reactor

A flexible reaction system that can be configured to suit your budget and application

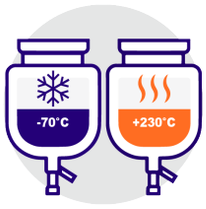
Reactor-Ready Flex is designed as a universal reactor work station, with a range of easily interchangeable vessels from 100 ml to 5 litres, which can be configured to suit the chemistry and scale needed for each project.

Vessels 100 ml to 5 Litres

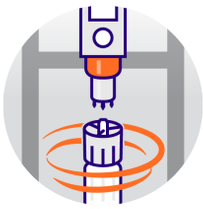
Rapid vessel exchange



Temp: -70ºC to +230ºC



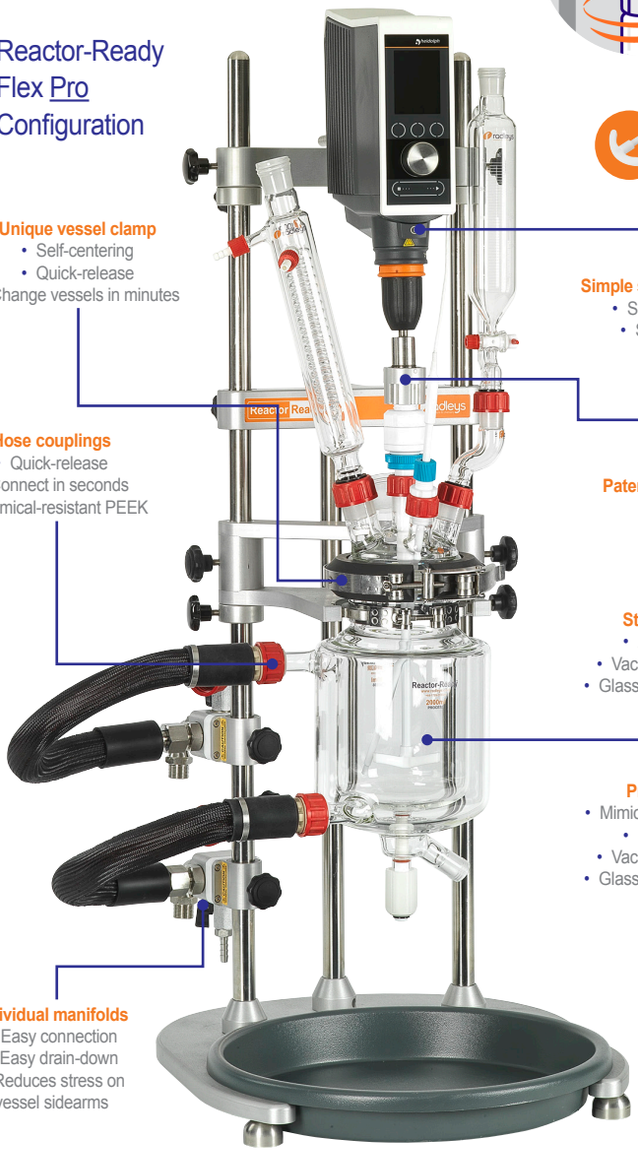
Self-aligning stirrer



Quick vessel drain down

Automation: Log & Control

Reactor-Ready Flex Pro Configuration



-70˚C 230˚C

Control and log your chemistry with AVA Lab Control Software

ava lab control software

Unique vessel clamp
- Self-centering
- Quick-release
- Change vessels in minutes

Simple stirrer alignment
- Slide and glide
- Set and lock

Hose couplings
- Quick-release
- Connect in seconds
- Chemical-resistant PEEK

Patented stirrer coupling
- Drop in
- No tools
- No fuss

Standard vessels
- 250 ml to 3 litres
- Vacuum jacketed option
- Glass accessories to match

Process vessels
- Mimic plant scale geometry
- 100 ml to 5 litres
- Vacuum jacketed option
- Glass accessories to match

Individual manifolds
- Easy connection
- Easy drain-down
- Reduces stress on vessel sidearms

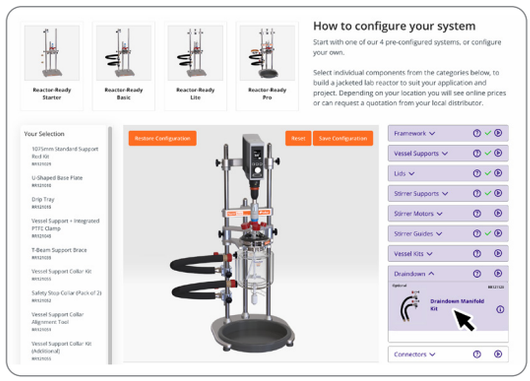
Convenient vessel kits make buying vessels and accessories easy and cost effective

A choice of popular stirrer paddles

Reactor-Ready Flex™ Lab Reactor

A modular jacketed lab reactor that accepts vessels from 100 ml to 5 litres, which can be swapped in minutes.

- Use one fumehood, circulator and overhead stirrer to save yourself time, space and money.
- Perfect for busy labs working on multiple projects. Select from a range of components and create a system with just the features you need.
- Start with one of our four pre-configured systems from an entry level system for a tight budget to a feature rich system for demanding applications, or configure your own.
- Upgrade any feature at any time for a truly flexible solution.
- You can automate any configuration with AVA Lab Control Software to control and data log your reactions.



Use our online configurator to create the perfect reactor for you:
https://www.radleys.com/rr-flex-configurator

Build your own, or start with one of our pre-configured systems...

Reactor-Ready Flex
Starter

Entry level Starter configuration where your budget may be limited

- Upgrade any of the components at any time
- Rectangular base
- Basic vessel support
- Metal vessel clamp
- Compact stirrer support
- PEEK fittings that connect directly to vessel sidearms
- Basic PTFE stirrer guide

Reactor-Ready Flex
Basic

The Basic configuration has the draindown manifolds added and features the popular PTFE clamp

- Rectangular base
- Basic vessel support
- PTFE vessel clamp
- Compact stirrer support
- Draindown manifold and hoses
- Basic PTFE stirrer guide

Reactor-Ready Flex
Lite

The Lite also features the versatile I-Beam and patented self aligning stirrer guide

- Rectangular base
- Basic vessel support
- PTFE vessel clamp
- I-Beam stirrer support
- Draindown manifold and hoses
- Self-aligning stirrer guide

Reactor-Ready Flex
Pro

The Pro is our original and most popular configuration featuring all the premium component options

- U-Shaped base with drip tray
- Vessel support with integrated PTFE clamp
- I-Beam stirrer support
- Draindown manifold and hoses
- Self-aligning stirrer guide

Reactor-Ready™ Filter Lab Reactor

-30°C 180°C

Reaction and filtration in a single jacketed vessel setup, ideal for crystallisation, reaction workups and product isolation

Reactor-Ready Filter Lab Reactor combines a jacketed lab reactor and a filter into one system. This allows users to isolate their product without having to manually transfer their material from reactor to a separate filter, saving time and reducing product losses.

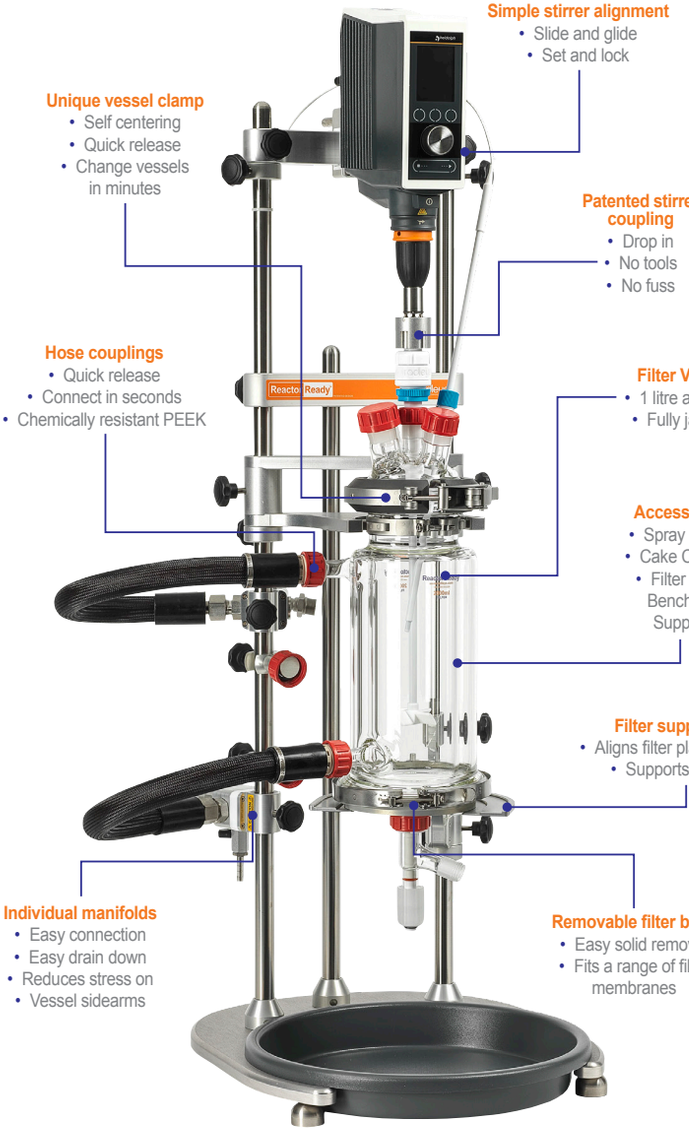
Features

- Filter vessels are fully jacketed down to base for accurate temperature control.
- Choice of jacketed filter vessels in 1 and 2 litre volumes.
- Temperature range of filter vessels: -30 ºC to +180 ºC (jacket temperature).
- Rapid vessel exchange with quick-release vessel clamp and hose couplings.
- Also accepts full range of Reactor-Ready Standard and Process vessels.
- Innovative hose manifolds allow easy thermofluid drain down.
- Accepts all leading brands of overhead stirrer for agitated slurrying and filtrations.
- Filter base accepts pre-cut filter membranes or you can cut your own membranes to size.
- Filter support plate aligns the base with the vessel for tool-free assembly.

Convert your Reactor-Ready Flex to a Filter Reactor

Control and log your chemistry with AVA Lab Control Software

ava lab control software



Simple stirrer alignment
- Slide and glide
- Set and lock

Unique vessel clamp
- Self centering
- Quick release
- Change vessels in minutes

Patented stirrer coupling
- Drop in
- No tools
- No fuss

Hose couplings
- Quick release
- Connect in seconds
- Chemically resistant PEEK

Filter Vessels
- 1 litre and 2 litre
- Fully jacketed

Accessories
- Spray Lance
- Cake Catcher
- Filter Plate Benchtop Support

Filter support plate
- Aligns filter plate with vessel
- Supports filter cake

Individual manifolds
- Easy connection
- Easy drain down
- Reduces stress on
- Vessel sidearms

Removable filter base
- Easy solid removal
- Fits a range of filter membranes

Convenient filter vessel kits make ordering vessels and accessories easy and cost effective

Filter vessels are offered with a turbine impeller

Choose from 1 or 2 litre, jacketed filter vessels

The framework is also compatible with the full range of Reactor-Ready Flex vessels from 100 ml to 5 litre

1 Litre Filter Vessel

2 Litre Filter Vessel

Reactor-Ready™ Duo Lab Reactor - *100 ml to 5 litres*

All the benefits of Reactor-Ready with two vessels in parallel or series

Reactor-Ready Duo shares the same unique features as Reactor-Ready, but holds two independent jacketed glass reaction vessels. The system can be configured to operate with a single thermoregulator controlling the jacket temperature of both vessels simultaneously or with two thermoregulators controlling the temperature of each vessel independently.

Features

-70°C 230°C

- Rapid exchange of both vessels independently, with quick-release vessel clamp and hose couplings.
- Choice of manifold kits allow two vessels to run from a single thermoregulator or two separate thermoregulators.
- System accepts two overhead stirrers which can be moved independently.

Applications

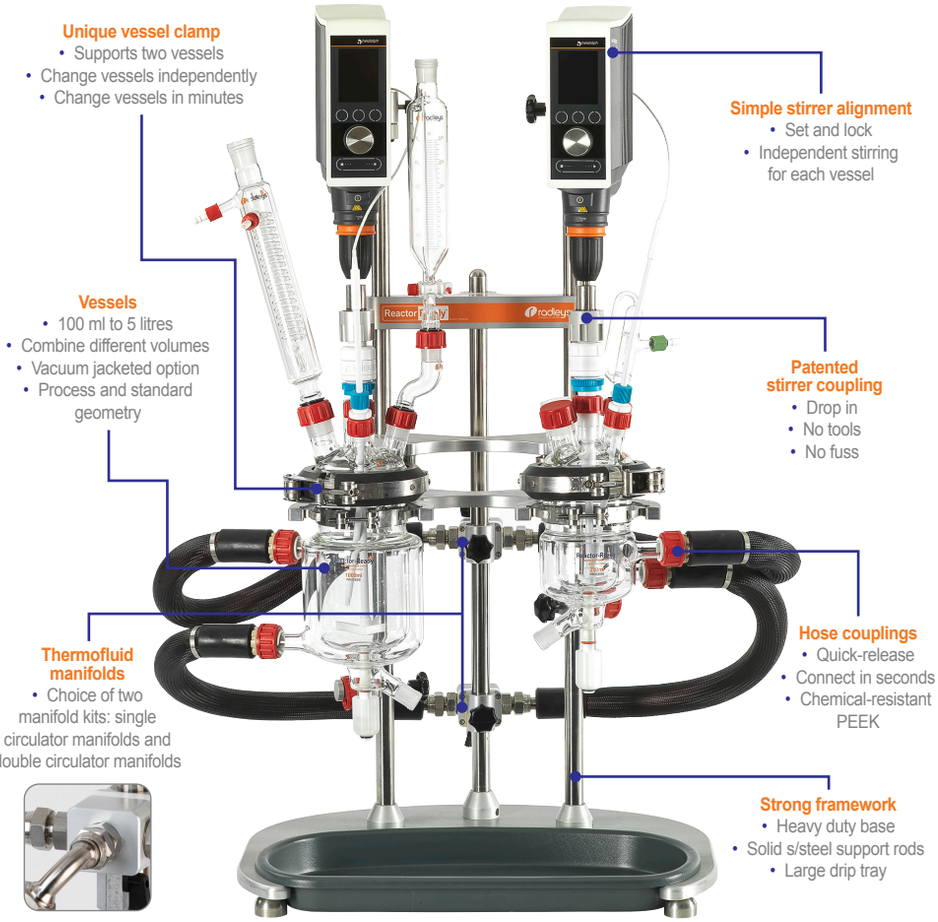
- Parallel synthesis or reaction optimisation: use similar or different size vessels and vary stirring speed, stirrer shape and temperature between vessels.
- Two stage reaction: transfer reactant from one vessel to the other using vacuum or a pump.
- Single reaction vessel: using the second vessel as either a receiving or feed vessel (where reagents can be pre-heated or pre-cooled prior to addition).
- Use optional AVA software to control fluid transfer between vessels.

Process vessels mimic the geometry of plant scale reactors, with a 1.25 to 1 ratio

1000 ml Process

1000 ml Standard

Control and log your chemistry with AVA Lab Control Software



Unique vessel clamp
- Supports two vessels
- Change vessels independently
- Change vessels in minutes

Simple stirrer alignment
- Set and lock
- Independent stirring for each vessel

Vessels
- 100 ml to 5 litres
- Combine different volumes
- Vacuum jacketed option
- Process and standard geometry

Patented stirrer coupling
- Drop in
- No tools
- No fuss

Thermofluid manifolds
- Choice of two manifold kits: single circulator manifolds and double circulator manifolds

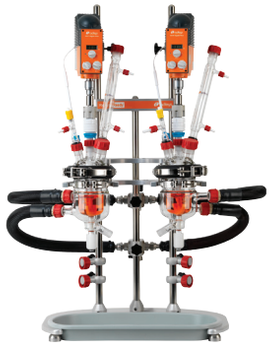
Hose couplings
- Quick-release
- Connect in seconds
- Chemical-resistant PEEK

Strong framework
- Heavy duty base
- Solid s/steel support rods
- Large drip tray

Supports two glass reaction vessels of different or similar volume from 100 ml to 5 litres

Similar vessel sizes with double circulator manifold kit

Different vessel sizes with double circulator manifold kit



Similar vessel sizes with single circulator manifold kit

Different vessel sizes with single circulator manifold kit

How to order Reactor-Ready Pilot

1. Select the Reactor-Ready Pilot Core.

2. Choose the lid you require (custom options available).

3. Choose the Vessel Kits you need.

4. Select the Lower Support Plate if using the 30 or 35 litre vessels

5. Select the overhead stirrer you need.

6. If you need a thermoregulator, hoses, hose adapters, thermofluid or accessory glassware, then select from the accessory list.

7. If you need automation add the AVA Lab Control Software and Data Hub.

ava Level 4 lab control software

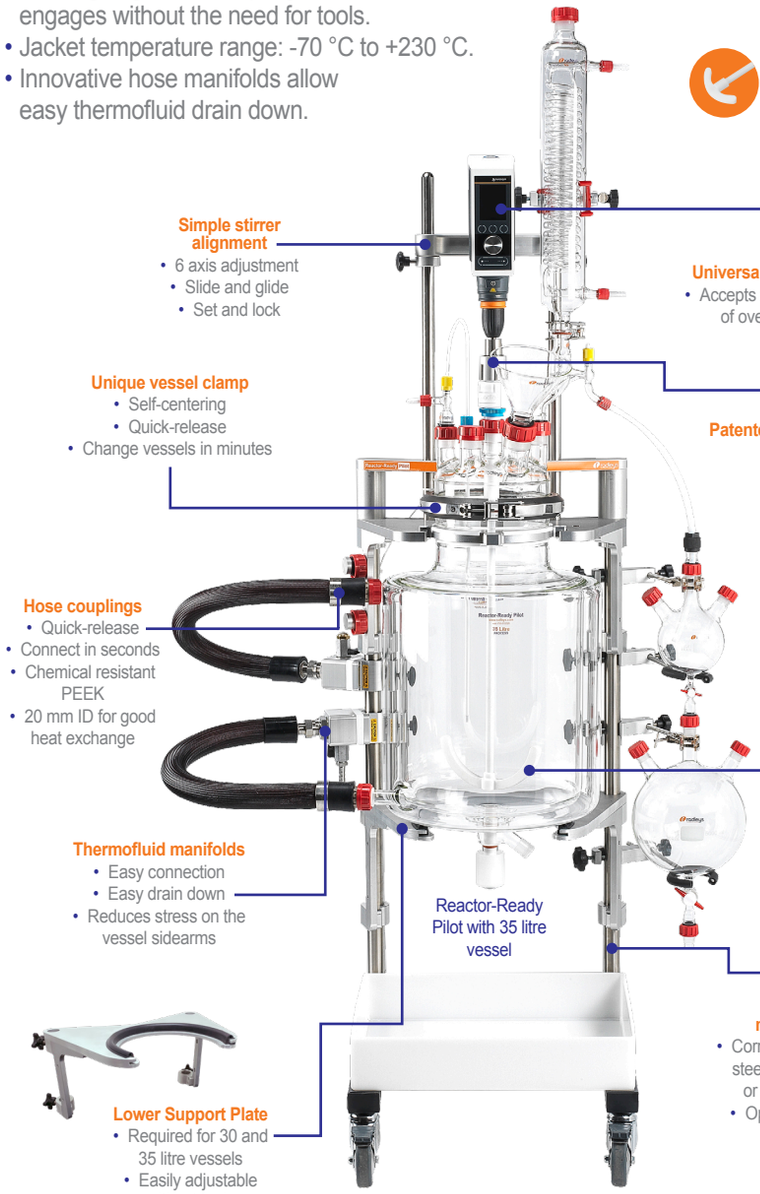
Reactor-Ready™ Pilot - *5 to 35 litres*

Replace multiple reactor set-ups with a single, universal pilot scale system with interchangeable vessels that can be swapped in minutes, not hours

Reactor-Ready Pilot is ideal for process development, scale-up, pilot and kilo labs.

Features

- Rapid vessel exchange with quick-release vessel clamp and wide bore hose couplings.
- Single jacketed vessels available in 5, 10, 15, 20, 30 and 35 litres.
- Vacuum jacketed vessels available in 5 and 10 litres.
- Vessels have 1.25 to 1 ratio of internal height to diameter to mimic plant scale reactors.
- DN200 vessel flange.
- Accepts all leading brands of overhead stirrer and allows easy, tool-free adjustment.
- Compact stainless steel framework accepts all vessel sizes.
- Self-aligning stirrer coupling engages without the need for tools.
- Jacket temperature range: -70 °C to +230 °C.
- Innovative hose manifolds allow easy thermofluid drain down.



Simple stirrer alignment
- 6 axis adjustment
- Slide and glide
- Set and lock

Universal stirrer support
- Accepts all leading brands of overhead stirrer

Unique vessel clamp
- Self-centering
- Quick-release
- Change vessels in minutes

Patented stirrer coupling
- Drop in
- No tools
- No fuss

Standard jacketed vessels
- Single jacketed vessels 5, 10, 15, 20, 30 and 35 litres
- Vacuum jacketed vessels 5 and 10 litres
- Zero dead space bottom outlet valve
- Ratio 1.25:1

Hose couplings
- Quick-release
- Connect in seconds
- Chemical resistant PEEK
- 20 mm ID for good heat exchange

Custom vessels
- To your specific size and style
- In-house manufacturing
- Mimic plant scale geometry

Thermofluid manifolds
- Easy connection
- Easy drain down
- Reduces stress on the vessel sidearms

Reactor-Ready Pilot with 35 litre vessel

Compact, mobile framework
- Corrosion-resistant stainless steel, anodised aluminium or powder-coated steel
- Option for wheels or feet

Lower Support Plate
- Required for 30 and 35 litre vessels
- Easily adjustable without tools

5 litre vacuum jacketed vessel

35 Litre Single jacketed vessel

Reactor-Ready Pilot with 20 litre jacketed vessel

AVA™ Software - *take control of your chemistry*

AVA Level 4 controls multi-device jacketed reaction systems

- Unattended chemistry for improved productivity
- Automatically log your process data
- Improve safety and reduce manual errors
- Control any reactor

ava level 4 lab control software

If you answer YES to any of the following:

Do you want to...

- ☑ Manage multiple devices easily during complex reactions?
- ☑ Have freedom from manual recording of experimental data?
- ☑ Safely run and monitor unattended reactions 24/7?
- ☑ Reduce manual errors or inconsistencies in your chemistry?
- ☑ Safely control exotherms?

Then AVA Software is what you need to take control of your chemistry...

Control multi-device reactions

- Control up to 4 reaction systems with up to 16 devices on one screen
- Create complex experiments with any number of steps in series or in parallel
- Pre-program multi-step recipes, with the flexibility to make and track on-the-fly adjustments
- Interlink devices and set feedback/control loops, end-point conditions and safety limits

No more manual data logging or manual errors

- Automatically record reaction parameters and log what you do, as you do it
- Repeat experiments accurately for reproducible and consistent results
- Create reports in a few clicks or export data as a CSV file for further analysis
- Share results between users to improve research and collaboration

Safe unattended chemistry

- Automate cooling during exothermic events
- Link devices such as balances and pumps for controlled, unattended reagent addition
- Define safety overrides and cut-off conditions
- Configure audio and visual alarms

Download and try AVA software for FREE

Download Free Demo www.radleys.com

Learn how AVA software works - try before you buy

- Simulate control of devices
- Set up apparatus and control experimental Schedules
- Share setups and Schedules with other AVA users
- Analyse results and create reports for real or simulated data
- Find out more about AVA Level 1-3 on our website

AVA Care Support

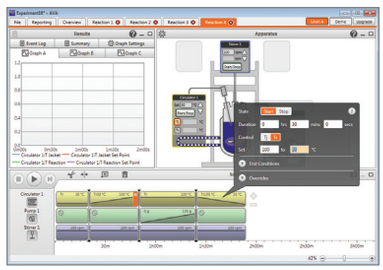
- Free support for 1st year
- Free priority email and telephone support
- Free set-up and application support
- Free software updates during support period, keeping software current.

Data Hub

Integrates devices via an RS232 serial interface

Level 4

Control and log reaction systems with multiple devices



Control a variety of devices

AVA software includes a library of pre-configured driver files allowing easy integration with a wide range of 3rd party devices.

Radleys also provide a 'New Driver Configuration Service' if required.

Accelerating chemistry

Our amazing team of international distributors at our distributors awards event

Radleys provide innovative productivity tools helping chemists discover amazing things.
Chemistry equipment for safer, cleaner, greener and more productive chemical research.

Visit **www.inkarp.co.in** to see our full range of chemistry productivity tools.

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Inkarp Instruments Pvt Ltd (Head Office)

Plot No. 5A/10-11, 3rd Floor, IDA Nacharam Road No. 1, Nacharam - Chilka Nagar Road, Hyderabad - 500076

Email: sales@inkarp.co.in | service@inkarp.co.in | Ph. No.: +91 99666 34008 | Telephone: +91 40 2717 2293

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